

Introduction

Formula 1 is a sport of precision, where every decision can make the difference between victory and defeat. Beyond speed and skill, **data science** plays a crucial role in understanding race outcomes.

This study explores how **starting grid position**, **weather conditions**, and **pit stop time** influence race results.



How does the starting grid position influence the finishing position of drivers in the seasons from 1994 - 2024

The starting grid position is determined by the qualifying session the day before, but how beneficial is the starting position?

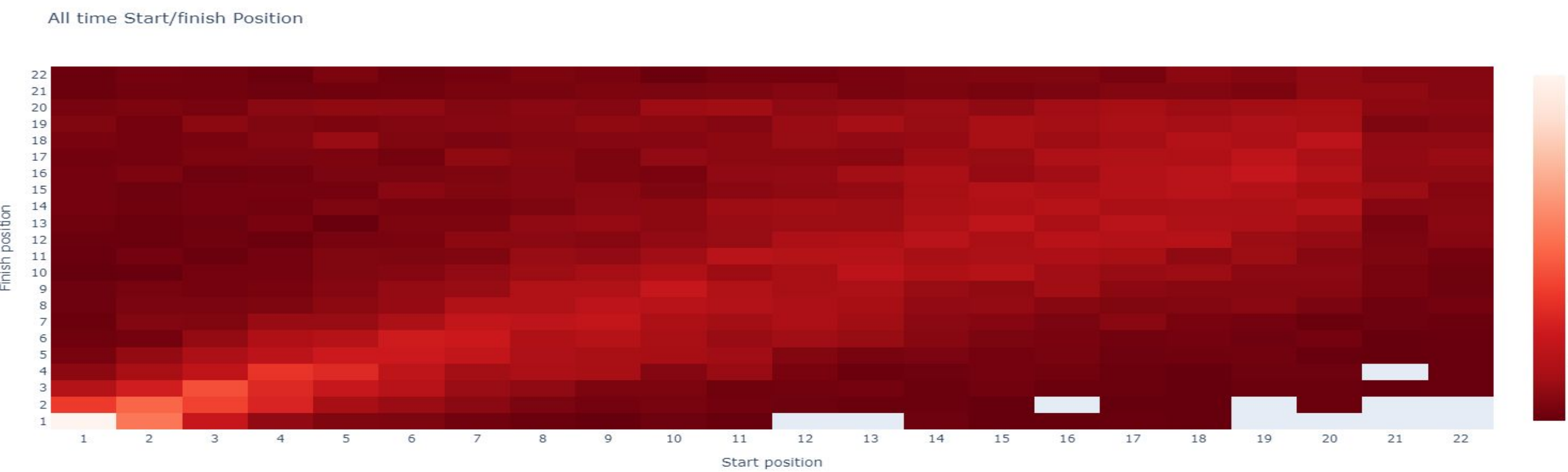


Figure 1: Start and finish position from 1994 - 2024

- ➔ Highest chance to win a race from pole position
- ➔ Finish position tends to fall within a 2-place range around the starting position.

How does the number and the average duration of pit stops for a driver in a race relate to his finishing position?

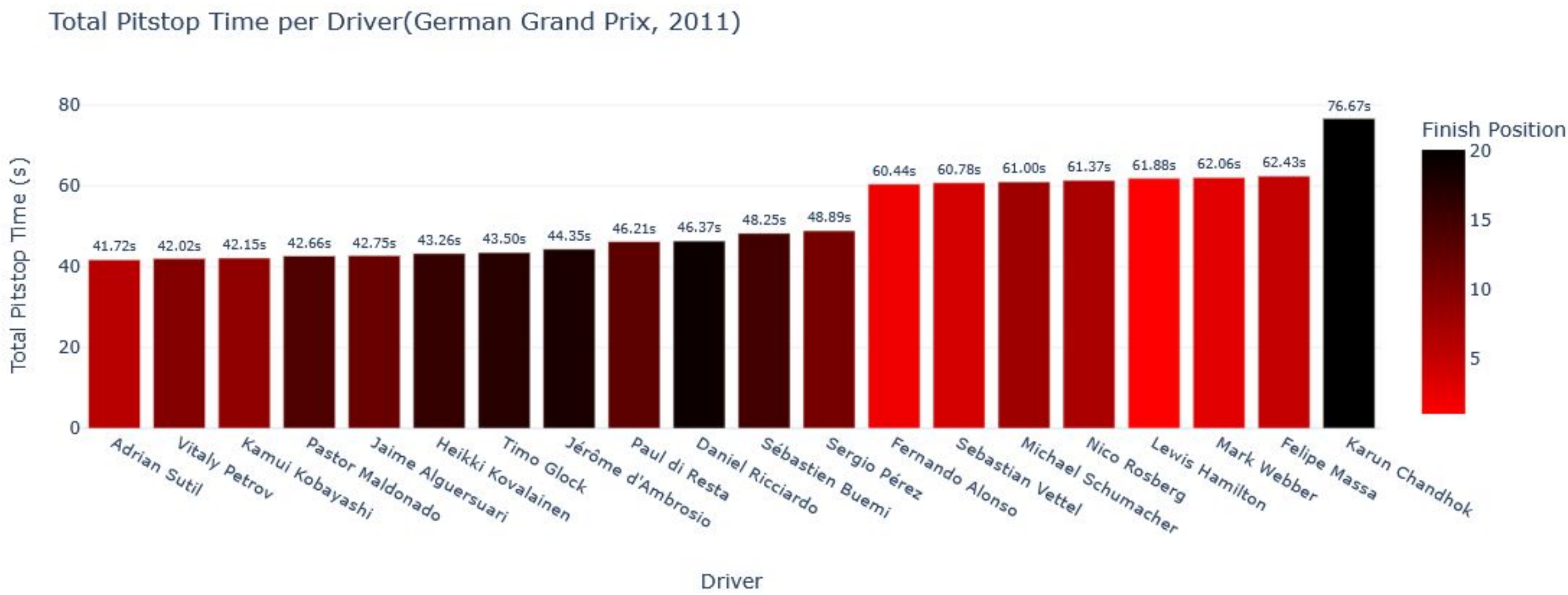


Figure 2 : Total pit stop time and finishing position (German Grand Prix, 2012)

Pit stops are essential for tire changes and adjustments. This visualization compares driver standings with total pit stop time. Based on the data, no clear relationship between time spent on pit stops and final positions is observable.

How does weather affect race completion rates and the likelihood of crashes or retirements in the seasons from 2005 - 2024?

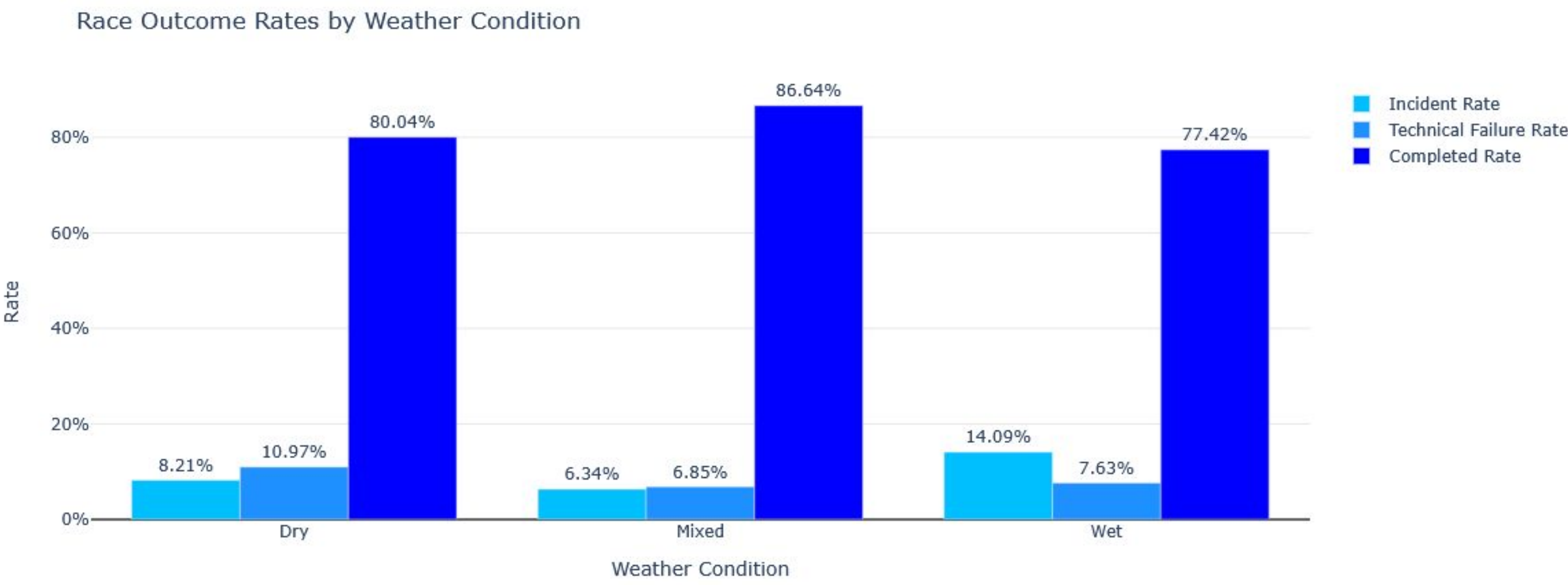


Figure 3 : Race Outcome by Weather Conditions

When a race ends early for a driver, it is called a retirement. A **retirement** can be caused by an **incident** or a **technical failure**. Other reasons are excluded. It is observable that:

- ➔ With **wet weather**, compared to dry and mixed weather, there is a significantly **higher incident rate**
- ➔ The technical failure rate is lower during mixed and wet weather compared to dry weather.
- ◆ With wet weather the driving speed is usually lower, so the car is less stressed.

How has the total number of crashes and retirements evolved in the seasons from 1994 - 2024?

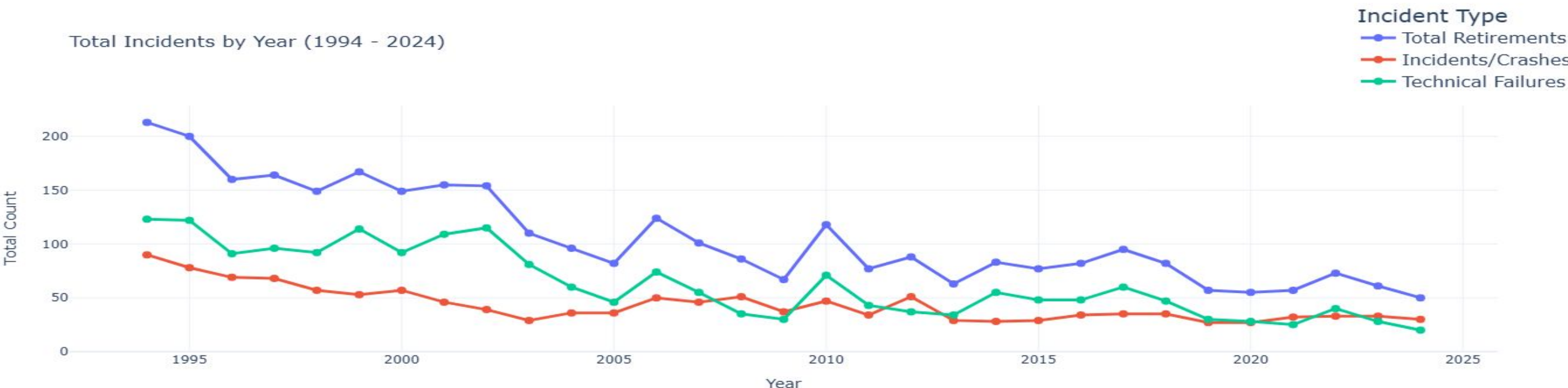
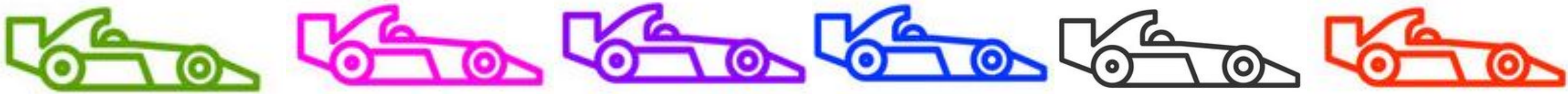


Figure 4 : Total Incidents by Year (1994 - 2024)

- ➔ The amount of Retirements has been in decline since 1994
- ➔ Technical failures were the leading cause of retirements until about 2005.
- ➔ Spikes in 2006 and 2010 coincided with major regulation changes and increases in drivers.
- ➔ Crash-related retirements declined from 90 in 1994 to about 30 per season since 2013.

For more information check our website by following the cars below !



Sources

[1] Jolpica API Abrufdatum 18.03.2025 URL: <https://github.com/jolpica/jolpica-f1/tree/main>
[2] Open meteo API Abrufdatum 18.03.2025 URL: <https://open-meteo.com/en/docs/historical-weather-api>